LDRP Institute of Technology and Research

MCA Department

Subject Name: - Advance Database Management System

Practical Assignment: 2

Name : Jivani Riya Aminbhai

Roll No : 13

Div : A

1	**Write a PL/SQL block to display "Welcome to PL/SQL Programming".**
Code:	SQL> SET SERVEROUTPUT ON; SQL> BEGIN 2 DBMS_OUTPUT.PUT_LINE('Welcome to PL/SQL Programming'); 3 END; 4 / Welcome to PL/SQL Programming
Output:	Welcome to PL/SQL Programming PL/SQL procedure successfully completed.
2	**Write a PL/SQL block to declare variables for student name, age, course and city. Assign suitable values and display the information.**
Code:	SQL> DECLARE 2 student_name VARCHAR2(30); 3 age NUMBER;

	4 course VARCHAR2(20); 5 city VARCHAR2(20); 6 BEGIN 7 student_name := 'Riya'; 8 age := 20; 9 course := 'MCA'; 10 city := 'Gandhinagar'; 11 12 DBMS_OUTPUT.PUT_LINE('Student Name: ' || student_name); 13 DBMS_OUTPUT.PUT_LINE('Age: ' || age); 14 DBMS_OUTPUT.PUT_LINE('Course: ' || course); 15 DBMS_OUTPUT.PUT_LINE('City: ' || city); 16 END; 17 /
Output:	Student Name: Riya Age: 20 Course: MCA City: Gandhinagar PL/SQL procedure successfully completed.
3	**Write a PL/SQL block to declare two numbers, calculate their sum and display the result.**
Code:	**SQL> DECLARE** **2 num1 NUMBER;** **3 num2 NUMBER;** **4 total NUMBER;**

	5 BEGIN **6 num1 := 10;** **7 num2 := 20;** **8** **9 total := num1 + num2;** **10** **11 DBMS_OUTPUT.PUT_LINE('First Number: ' || num1);** **12 DBMS_OUTPUT.PUT_LINE('Second Number: ' || num2);** **13 DBMS_OUTPUT.PUT_LINE('Sum: ' || total);** **14 END;** **15 /**
Output:	First Number: 10 Second Number: 20 Sum: 30 PL/SQL procedure successfully completed.
4	**Write a PL/SQL block to perform addition, subtraction, multiplication and division on two numbers.**
Code:	SQL> DECLARE 2 num1 NUMBER; 3 num2 NUMBER; 4 addition NUMBER; 5 subtraction NUMBER; 6 multiplication NUMBER; 7 division NUMBER; 8 BEGIN 9 num1 := 20;

	10 num2 := 10; 11 12 addition := num1 + num2; 13 subtraction := num1 - num2; 14 multiplication := num1 * num2; 15 division := num1 / num2; 16 17 DBMS_OUTPUT.PUT_LINE('Addition: ' || addition); 18 DBMS_OUTPUT.PUT_LINE('Subtraction: ' || subtraction); 19 DBMS_OUTPUT.PUT_LINE('Multiplication: ' || multiplication); 20 DBMS_OUTPUT.PUT_LINE('Division: ' || division); 21 END; 22 /
Output:	Addition: 30 Subtraction: 10 Multiplication: 200 Division: 2 PL/SQL procedure successfully completed.
5	**Write a PL/SQL block to store marks of three subjects and calculate the total and average marks.**
Code:	SQL> DECLARE 2 marks1 NUMBER; 3 marks2 NUMBER; 4 marks3 NUMBER; 5 total NUMBER; 6 average NUMBER;

	7 BEGIN 8 marks1 := 80; 9 marks2 := 75; 10 marks3 := 85; 11 12 total := marks1 + marks2 + marks3; 13 average := total / 3; 14 15 DBMS_OUTPUT.PUT_LINE('Subject 1 Marks: ' || marks1); 16 DBMS_OUTPUT.PUT_LINE('Subject 2 Marks: ' || marks2); 17 DBMS_OUTPUT.PUT_LINE('Subject 3 Marks: ' || marks3); 18 DBMS_OUTPUT.PUT_LINE('Total Marks: ' || total); 19 DBMS_OUTPUT.PUT_LINE('Average Marks: ' || average); 20 END; 21 /
Output:	Subject 1 Marks: 80 Subject 2 Marks: 75 Subject 3 Marks: 85 Total Marks: 240 Average Marks: 80 PL/SQL procedure successfully completed.
6	**Write a PL/SQL block to calculate the area and perimeter of a rectangle using length and width.**
Code:	SQL> DECLARE 2 length NUMBER; 3 width NUMBER;

	4 area NUMBER; 5 perimeter NUMBER; 6 BEGIN 7 length := 10; 8 width := 5; 9 10 area := length * width; 11 perimeter := 2 * (length + width); 12 13 DBMS_OUTPUT.PUT_LINE('Length: ' || length); 14 DBMS_OUTPUT.PUT_LINE('Width: ' || width); 15 DBMS_OUTPUT.PUT_LINE('Area: ' || area); 16 DBMS_OUTPUT.PUT_LINE('Perimeter: ' || perimeter); 17 END; 18 /
Output:	Length: 10 Width: 5 Area: 50 Perimeter: 30 PL/SQL procedure successfully completed.
7	**Write a PL/SQL block to calculate simple interest using principal amount, rate and time.**
Code:	SQL> DECLARE 2 principal NUMBER; 3 rate NUMBER;

	4 time NUMBER; 5 simple_interest NUMBER; 6 BEGIN 7 principal := 10000; 8 rate := 5; 9 time := 2; 10 11 simple_interest := (principal * rate * time) / 100; 12 13 DBMS_OUTPUT.PUT_LINE('Principal Amount: ' || principal); 14 DBMS_OUTPUT.PUT_LINE('Rate: ' || rate || '%'); 15 DBMS_OUTPUT.PUT_LINE('Time: ' || time || ' years'); 16 DBMS_OUTPUT.PUT_LINE('Simple Interest: ' || simple_interest); 17 END; 18 /
Output:	Principal Amount: 10000 Rate: 5% Time: 2 years Simple Interest: 1000 PL/SQL procedure successfully completed.
8	**Write a PL/SQL block to store a student's name, age, course and city and display the information using the concatenation operator ||.**
Code:	SQL> DECLARE 2 student_name VARCHAR2(30);

	3 age NUMBER; 4 course VARCHAR2(20); 5 city VARCHAR2(20); 6 BEGIN 7 student_name := 'Riya'; 8 age := 20; 9 course := 'MCA'; 10 city := 'Gandhinagar'; 11 12 DBMS_OUTPUT.PUT_LINE('Student Name: ' || student_name || 13 ', Age: ' || age || 14 ', Course: ' || course || 15 ', City: ' || city); 16 END; 17 /
Output:	Student Name: Riya, Age: 20, Course: MCA, City: Gandhinagar PL/SQL procedure successfully completed.
9	**Write a PL/SQL block to store a student's date of birth and display it along with the current date using SYSDATE.**
Code:	SQL> DECLARE 2 dob DATE;

	3 BEGIN 4 dob := TO_DATE('01-JUL-2006', 'DD-MON-YYYY'); 5 6 DBMS_OUTPUT.PUT_LINE('Date of Birth: ' ‖ TO_CHAR(dob, 'DD-MON-YYYY')); 7 DBMS_OUTPUT.PUT_LINE('Current Date: ' ‖ TO_CHAR(SYSDATE, 'DD-MON-YYYY')); 8 END; 9 /
Output:	Date of Birth: 01-JUL-2006 Current Date: 26-AUG-2026 PL/SQL procedure successfully completed.
10	**Write a PL/SQL block to store basic salary, allowance and deduction of an employee and calculate the final salary.**
Code:	SQL> DECLARE 2 basic_salary NUMBER; 3 allowance NUMBER; 4 deduction NUMBER; 5 final_salary NUMBER; 6 BEGIN 7 basic_salary := 30000; 8 allowance := 5000; 9 deduction := 2000;

	10 11 final_salary := basic_salary + allowance - deduction; 12 13 DBMS_OUTPUT.PUT_LINE('Basic Salary: ' || basic_salary); 14 DBMS_OUTPUT.PUT_LINE('Allowance: ' || allowance); 15 DBMS_OUTPUT.PUT_LINE('Deduction: ' || deduction); 16 DBMS_OUTPUT.PUT_LINE('Final Salary: ' || final_salary); 17 END; 18 /
Output:	Basic Salary: 30000 Allowance: 5000 Deduction: 2000 Final Salary: 33000 PL/SQL procedure successfully completed.